

李金涛

lijintao@zju.edu.cn

<https://jintaolee-roger.github.io>

浙江大学，中国浙江杭州

工作经历

2025–至今 博士后研究人员，浙江大学
合作导师：李伟昌教授（长江学者讲座教授）

教育经历

2020–2025 博士，地球物理学，中国科学技术大学
导师：伍新明教授（国家杰青/青 A）
学位论文：《基于深度学习的高分辨率叠后地震数据处理及应用》

2016–2020 学士，地球物理学，中国科学技术大学

研究方向

- 地球科学人工智能与基础模型
- 地震数据处理、解释与反演
- DAS、光纤感知、多传感器融合与地球物理监测
- 地球科学智能基础设施与开源软件

科研基金

2027–2029 国家青年科学基金项目（C 类）[原青年科学基金项目]
主持；资助金额：30 万元；执行期：2027.01–2029.12

2026–2028 中国博士后科学基金第 79 批面上资助
主持；资助编号：2026M790497；资助金额：8 万元；执行期：2026.09–2028.08

2026–2028 国家资助博士后研究人员计划 B 档
主持；资助编号：GZB20260111

论文发表

Google Scholar

* → 共同第一作者, † → 通讯作者

已发表成果

1. Wang, Yuting, **Jintao Li**[†], Xiaoming Sun, and Xinming Wu[†]. 2026. “Deep Learning-based Seismic Reflectivity Estimation by Pre-training on Labeled Synthetic Data and Physics-guided Fine-tuning in Field Data.” *Geophysical Journal International (GJI)*, ggag317. <https://doi.org/10.1093/gji/ggag317>.
2. **Li, Jintao**, Xinming Wu, Xianwen Zhang, Xin Du, Xiaoming Sun, Bao Deng, and Guangyu Wang. 2026. “High-Fidelity Seismic Super-Resolution Using Prior-Informed Deep Learning with 3D Awareness.” *IEEE Transactions on Image Processing (IEEE TIP)* 35:2152–2166. <https://doi.org/10.1109/TIP.2026.3663926>.
– 中科院一区 TOP, CCF-A, IF: 15.3。
3. **Li, Jintao**, and Xinming Wu. 2026. “Memory-Efficient Full-Volume Inference for Large-Scale 3D Dense Prediction without Performance Degradation.” *Communications Engineering* 5:20. <https://doi.org/10.1038/s44172-025-00576-2>.
– Nature Portfolio 期刊, IF: 7.0
– 该论文获得三位审稿人的一致认可。
4. **Li, Jintao**, Yunzhi Shi, and Xinming Wu. 2025. “CIGVis: An Open-source Python Tool for the Real-time Interactive Visualization of Multidimensional Geophysical Data.” *Geophysics* 90 (1): F1–F10. <https://doi.org/10.1190/geo2024-0041.1>.
5. **Li, Jintao**, Xinming Wu, Yueming Ye, Cun Yang, Zhanxuan Hu, Xiaoming Sun, and Tao Zhao. 2023. “Unsupervised Contrastive Learning for Seismic Facies Characterization.” *Geophysics* 88 (1): WA81–WA89. <https://doi.org/10.1190/geo2022-0148.1>.
6. **Li, Jintao**, Xinming Wu, and Zhanxuan Hu. 2022. “Deep Learning for Simultaneous Seismic Image Super-Resolution and Denoising.” *IEEE Transactions on Geoscience and Remote Sensing (IEEE TGRS)* 60:1–11. <https://doi.org/10.1109/TGRS.2021.3057857>.
– 中科院一区 TOP, ESI 高被引论文, IF: 9.4。
7. Gao, Hui, Xinming Wu, **Jintao Li**, Xiaoming Sun, and Jiarun Yang. 2026. “Massive-scale Unlabeled Field and Labeled Synthetic Seismic Datasets of Global Shelf-edge Clinothems.” *Geophysical Journal International (GJI)*, ggag295. <https://doi.org/10.1093/gji/ggag295>.
8. Sheng, Hanlin, Xinming Wu, Xu Si, **Jintao Li**, Sibao Zhang, and Xudong Duan. 2025. “Seismic Foundation Model: A Next Generation Deep-learning Model in Geophysics.” *Geophysics* 90 (2): IM59–IM79. <https://doi.org/10.1190/geo2024-0262.1>.

– ESI 高被引论文。

9. Han, Long, Xinming Wu, Zhanxuan Hu, **Jintao Li**, and Huijing Fang. 2024. “MAMCL: Multi-attributes Masking Contrastive Learning for Explainable Seismic Facies Analysis.” *Computers & Geosciences* 193:105731.

未发表成果

1. **Li, Jintao**, Weichang Li, Kai Tong, and Xaingyu Guo. 2026. *FiLark: A Streaming-first Software Framework for End-to-end Exploration, Annotation, and Algorithm Integration in Distributed Acoustic Sensing*. arXiv: 2605.20132. <https://arxiv.org/abs/2605.20132>.
– Submitted to *Computers & Geosciences*; under review.
2. Han, Long, Xinming Wu[†], Xiaoming Sun, **Jintao Li**, and Yimin Dou. 2026. *A Universal Diffusion Regularizer for Geophysical Inversion*.
– Under review.
3. Dou, Yimin, Xinming Wu, Nathan L. Bangs, Harpreet Singh Sethi, **Jintao Li**, Hang Gao, and Zhixiang Guo. 2026. *Geological Everything Model 3D: A Promptable Foundation Model for Unified and Zero-shot Subsurface Understanding*. arXiv: 2507.00419. <https://arxiv.org/abs/2507.00419>.
– In revision.
4. Sheng, Hanlin, Xinming Wu, Hang Gao, Haibin Di, Sergey Fomel, **Jintao Li**, and Xu Si. 2025. *On the Workflow, Opportunities and Challenges of Developing Foundation Models in Geophysics*. arXiv: 2504.17384. <https://arxiv.org/abs/2504.17384>.

其他会议论文与摘要

1. **Li, Jintao**, Yunzhi Shi, and Xinming Wu. 2024. “CIGVis: An Open-source Python Tool for Real-time Interactive Visualization of Multidimensional Geophysical Data.” In *85th EAGE Annual Conference & Exhibition*, 1–5. <https://doi.org/10.3997/2214-4609.2024101065>.
2. **Li, Jintao**, Xinming Wu, and Zhanxuan Hu. 2020. “Deep Learning for Simultaneous Seismic Super-Resolution and Denoising.” In *SEG International Exposition and Annual Meeting*. <https://doi.org/10.1190/segam2020-3426412.1>.
3. Sheng, Hanlin, Xinming Wu, Xu Si, **Jintao Li**, Sibao Zhang, and Xudong Duan. 2024. “Seismic Foundation Model (SFM): All-purpose Feature Extraction from Seismic Data for Diverse Geophysical Applications.” In *85th EAGE Annual Conference & Exhibition*, 1–5. <https://doi.org/10.3997/2214-4609.2024101033>.

4. Si, Xu, Xinming Wu, **Jintao Li**, Xin Cui, Hanlin Sheng, Hang Gao, Zefeng Li, and Zhi-gang Peng. 2024. “Tuning into Earth’s Voice: Leveraging Large Language Models for Earthquake Detection.” In *AGU Fall Meeting Abstracts*, S11B–3384. <https://ui.adsabs.harvard.edu/abs/2024AGUFMS11B.3384S/abstract>.

开源软件

- [cigvis](#): 多维地球物理数据交互式可视化工具。
- [cigsegy](#): SEG-Y 与 NumPy 数据转换及处理工具。
- [torchseis](#): 面向大规模三维地震数据推理的工具库。
- [SeismicSuperResolution](#): 地震图像超分辨率与去噪代码库。

荣誉

2025	中国科学技术大学优秀毕业生
2024	中石油奖学金
2021	研究生国家奖学金
2021–2025	中国科学技术大学一等学业奖学金
2020	中国科学技术大学优秀本科毕业论文奖
2019	国家励志奖学金

教学经验

中国科学技术大学

2023	专题讲师 ，《人工智能地球物理》 独立讲授约 3 学时“基于 Python 的科学数据可视化”专题。
2021	助教组长 ，《人工智能地球物理》 协调 8 名助教，参与课程作业与评分标准设计。

学术服务

会议服务

2026 年 11 月 技术委员会委员, SEG-GeoAI 2026 Workshop, 中国杭州

期刊审稿人

Geophysics; *IEEE TGRS*; *IEEE TSP*; *Sci. Rep.*; *Comput. Geosci.*; *JGR-MLC*; *IEEE GRSL*; *Interpretation*; *JGE*。

最后更新: 2026 年 9 月 7 日